

A The causal role recurs across all five Mamba-1 scales

	130M	370M	790M	1.4B	2.8B
Selected / control	P3 / P5	P3 / P5	P2 / P5	P5 / P4	P3 / P5
Causal role	Supported	Supported	Supported	Supported	Supported

Common scale-local contrast: selected restoration minus coefficient-matched control.

Historical endpoint labels and inferential-family details are retained in Appendix A.4.

B XG2 centered linear CKA

	130M	370M	790M	1.4B	2.8B
130M	1.00	0.74	0.44	0.34	0.56
370M	0.74	1.00	0.56	0.50	0.65
790M	0.44	0.56	1.00	0.66	0.64
1.4B	0.34	0.50	0.66	1.00	0.53
2.8B	0.56	0.65	0.64	0.53	1.00

C XG4 centered linear CKA

	130M	370M	790M	1.4B	2.8B
130M	1.00	0.54	0.45	0.48	0.54
370M	0.54	1.00	0.56	0.44	0.59
790M	0.45	0.56	1.00	0.44	0.39
1.4B	0.48	0.44	0.44	1.00	0.46
2.8B	0.54	0.59	0.39	0.46	1.00

Same 300 response-blind pairs per family. CKA = 1 denotes identical centered sample geometry;

off-diagonal values show partial, not invariant, cross-scale geometry. Cosine-RSM: Appendix A.2.